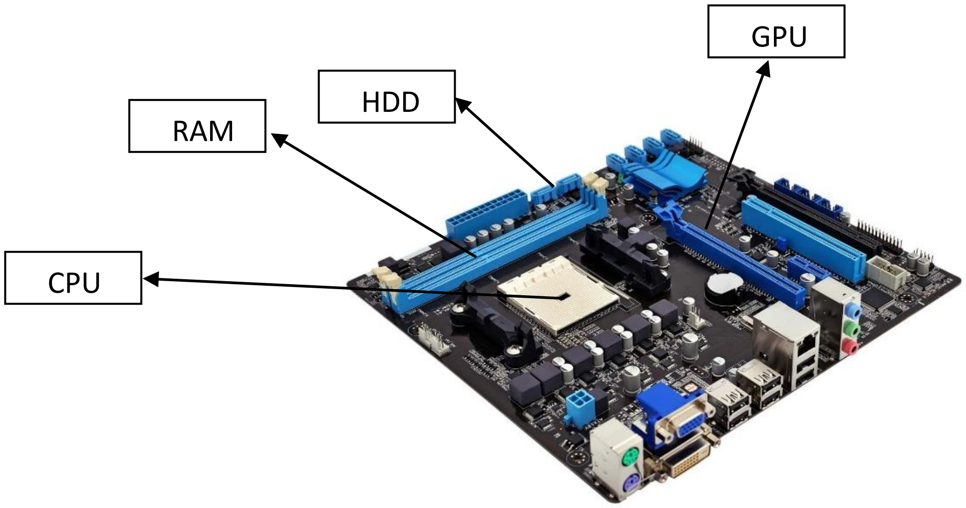
Q-1] Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



Q2] Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases

Hint- Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

	HDD	**SSD**
STANDS FOR	Hard Disk Drive.	Solid State Drive.
WORKING	Stores data on magnetic platters.	Stores files digitally on electronic circuits.
READING PROCESS	A signal is sent by the drive so a tiny arm is moved over the spinning disk, then the information is read.	The drive looks up for the data's exact location and digitally reads it.
WRITING PROCESS	It will find the nearest blank space on the disk then uses it reading arm to write data there.	It copies data to a new block, and then erases the old block. It then writes new to the old block by changing it
PERFORMANCE	It moves slower due to the continuous movement of platter , noisy and releases more heat.	It works faster, doesn't make noise , stays cool.

COST	Less Costly	More Costly
DURABILITY	They are less tough than SSD as it possess moving part inside hence they break more easily.	It is only a microchip so they are much harder to damage if you drop it accidentally.

Conclusion:
You should use a solid state drive (SSD) when you need high speeds or deal with frequent read/writes on large data volume. SSDs are a better choice for data analytics or gaming workloads.
On the other hand, a hard disk drive (HDD) is a better choice if you are dealing with data backups, data archives, or throughput-intensive workloads. SSDs are more cost-effective for storing high-volume data with infrequent access.

Q-3] Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU, RAM and GPU works as a team to run a game. game's logic is handled by the CPU, active information is easily kept accessible by RAM, proper visuals on screen is created or maintained by GPU.

<u>CPU [Central Processing Unit]</u>
It is the brain of a system that handles all the processing.
IT manages to player movements, handles the result such as a bulletr being fired or car being driven. Then sends all the calculated data to the GPU to be drawn.

<u>RAM [Random Access Memory]</u>
It is a temporary workspace that keeps required game information ready for the CPU.
Without waiting for the storage drive RAM load the map, all the needed game option so the CPU can access them instantly.
When we have enough RAM it ensures the game run smoothly nor freezes when loading new areas.

<u>GPU [Graphics Processing Unit]</u>
All the raw data from the CPU is taken and converted into the beautiful 3D graphics, colours, textures, etc you see on the monitor.
A powerful GPU allows you to play the game at higher resolutions and smoother frame rates.

Q-4]Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

If the PC is upgraded it means for the system not to lag and work properly. The following components needs to be upgraded.

GPU

It manages the graphics of the screen. Using the old card will make the game suffer as a result the player will have to play with the ugly settings. Hence upgrade is needed.

RAM

Temporarily data is held here so it can quickly load game's asset.
to run the latest version of FIFA smoothly atleast 8GB RAM is need but to work without hassle 12 GB to 16 GB is recommended for smooth, uninterrupted gameplay.

CPU

The "brain" of the computer that calculates where every player is running and what the physics of the ball are. A slow processor creates a "bottleneck," meaning your graphics card won't run at its full speed.